📘 MODULE 17: CIM (COMMON INFORMATION MODEL)

17.1 Concept

The **CIM** is a Splunk-defined **standard set of field names and data models** so that different log sources describe the same thing the same way.

Without CIM, a failed login is `user` in Linux, `Account_Name` in Windows, and `username` in a VPN. With CIM, all become `user`, `src`, `dest`, `action=failure`.

17.2 Why it matters to companies

- One detection works across all vendors.
- ES, dashboards, and `tstats` rely on CIM data models.
- New data sources just need **mapping**, and the detections work immediately.

It is a very common L2 and interview topic.

17.3 The mapping chain

```
RAW EVENT
 → sourcetype (parsing)
 → field extractions / aliases / calculated fields   (normalize field names + values)
 → eventtype   (saved search that identifies the event kind)
 → tag         (label such as "authentication")
 → DATA MODEL CONSTRAINT (tag=authentication)
 → event appears in the Authentication data model
```

17.4 Practical example of mapping

`eventtypes.conf`:

ini

```
[my_failed_login]
search = sourcetype=linux_secure "Failed password"
```

`tags.conf`:

ini

```
[eventtype=my_failed_login]
authentication = enabled
```

`props.conf`:

ini

```
[linux_secure]
EVAL-action = if(match(_raw,"Failed password"),"failure","success")
FIELDALIAS-src = src_ip AS src
```

- The **eventtype** identifies the events, and the **tag** puts them in the model.
- **EVAL-** (calculated field) creates `action`. **FIELDALIAS-** gives a second name.
- In real life, **vendor add-ons** (e.g. Splunk Add-on for Unix and Linux or for Microsoft Windows) ship all of this. You rarely write it from scratch, but you must be able to verify and fix it.

17.5 Normalized fields (Authentication example)

Field	Meaning
`action`	`success` or `failure`
`user`	Account being authenticated
`src`	Where the attempt came from
`dest`	System being logged into
`app`	Application involved
`signature`	Event name or ID

17.6 Important CIM data models (practical level)

Data model	Covers	Typical SOC use
Authentication	Logins, failures, privilege	Brute force, password spraying
Network Traffic	Firewall/flow records	Port scans, large transfers
Web	Proxy, web server	Malicious URLs, user agents
Endpoint	Processes, services, ports, filesystem	Suspicious process execution
Malware	AV detections	Infection tracking
Intrusion Detection	IDS/IPS alerts	Attack signatures
Email	Mail gateway	Phishing
Change	Account/config changes	Audit
Network Resolution (DNS)	DNS queries	C2 / DGA domains
Vulnerabilities	Scanner results	Exposure

17.7 Where the CIM lives

- Installed as the **Splunk Common Information Model Add-on** (free on Splunkbase, needs a splunk.com account). ES includes it.
- **Add-ons** map sourcetypes to CIM. The **CIM add-on** defines the models. Don't mix them up.

17.8 Validate that data is CIM-compliant

```spl
tag=authentication | stats count by sourcetype
```

Shows which sourcetypes are tagged as authentication.

```spl
| datamodel Authentication Authentication search
| stats count by sourcetype, action
```

Shows what the model actually contains. If `action` is empty or `sourcetype` is missing, the mapping is incomplete.

```spl
| tstats count FROM datamodel=Authentication BY Authentication.action
```

Use the CIM add-on's setup and data model audit pages to check acceleration and size.

17.9 Hands-on

1. Install the **Splunk Common Information Model** add-on and the **Splunk Add-on for Unix and Linux** (Apps → Find More Apps, or download from Splunkbase and use "Install app from file"). Restart.
2. Run `tag=authentication | stats count by sourcetype`.
3. In the CIM add-on's setup page, choose the indexes the models should search (e.g. `linux_auth`), then enable acceleration for `Authentication` only.
4. Run the Module 16 `tstats` query on `Authentication`.
5. Write down which required fields (`action`, `user`, `src`) are missing from your data.

17.10 Troubleshooting

Symptom	Cause
Data not in the model	Missing tag, eventtype doesn't match, or index not in the CIM allowed indexes
Fields empty (`user`, `src`)	Missing alias or extraction, or wrong sourcetype
`tstats` empty but raw search works	No acceleration, or `summariesonly=true` too early
Wrong values (`action=failed`)	Needs normalization to `failure` with a calculated field or lookup

17.11 Interview Q&A

- **What is CIM?** A standard schema of field names and data models for normalizing different data sources.
- **Why does it matter?** Detections, dashboards, and ES work across vendors without rewriting.
- **How do you make a new source CIM-compliant?** Correct sourcetype → extractions/aliases/calculated fields → eventtype → tag → verify it appears in the data model.
- **Add-on vs CIM app?** The add-on maps a vendor's data to CIM, and the CIM app defines the models.
- **Field alias vs calculated field?** An alias renames a field. A calculated field computes a new one with `eval`.

17.12 Revision

CIM = standard field names · chain: sourcetype → fields → eventtype → tag → data model · add-ons map, CIM app defines · validate with `tag=` and `datamodel`.